

Supplementary document

Hello and welcome to this exploration of our material world. Today, we are going to decorate the fascinating rules that absolutely govern everything around us. This is really the ideal starting point, especially in college, to understand the basics of matter. Come on, we plunge directly into the heart of the subject and discover how science observes and ranks everything that has a mass and takes place in the universe. Water and ice. What's crazy is that it's exactly the same substance. Chemically, nothing changes. But if we spill a glass of water, it's a disaster, it's going everywhere. While an ice cube, it remains wisely placed on the table. Why do they behave so radically differently? It's this little detail that makes you skate on a frozen lake in winter, but swim there in summer. And understanding that is the whole stake of the states of matter. Here's what we're gonna talk about today. We will first define what a property is, make the differences between physical and chemical, and then review solids, liquids and gases. Before we put it all together, in a small summary. Part one, the concept of ownership. Yes, to properly classify matter, you must already know what you are talking about when using this word in physics-chemistry. To make it simple, a property is kind of a personality trait of matter. It's a clue to recognize a material because of its appearance, the sensation it gives the touch or how it reacts to its environment. It's exactly like describing someone by their size or the color of their eyes without even needing to know their name. With matter, it's the same, we identify it with its properties. Part two, physical or chemical. These famous personality traits of matter are divided into two broad categories. This is a really crucial distinction. On the one hand, we have physical properties. This is all we can observe or measure without destroying the material. Its colour, its melting temperature, or precisely if it is solid, liquid or gaseous. On the other hand, we have the chemical properties, which are a little more extreme. It only reveals itself when matter becomes something completely new. Take the burning wood. Its flammability is chemical property. Once burned, it's not wood, it's ashes. But today, we will focus 100% on a great physical property, the state of matter. This brings us to our third part, the solid state, with its own form and volume. Let's see how he behaves. If you slip a single block of wood into a glass beaker, you notice something right now. The solid completely ignores the shape of the container. He doesn't collapse, he doesn't fill the corners. There's just one cube left. The golden rule here is that a solid has its own fixed shape. It's also seizable, which means you can catch it with your fingers. Above all, it is totally incompressible. Even if we press the cube with all its strength, its volume will not change by one millimetre. It is thanks to this incompressible and seizable side that large brick buildings can be built safely. Solid is tough. Come on, let's go to Part Four, liquid state. Here, the rules of the game change completely. If one observes a liquid, there is a super interesting phenomenon, whether it is water or oil. If you lean the glass on the side, the surface of the liquid always remains perfectly horizontal. No matter the angle, gravity does its job and imposes this ultra flat surface. And unlike our wooden cube, liquids have absolutely no clean shape. They flow and perfectly sweep the smallest contours of the container that accommodate them, whether they are large, thin or completely twisted. But be careful not to confuse form and volume. If you lock water in a syringe, close the end and press like a madman on the plunger, nothing happens, the liquid is stubborn, it refuses to shrink, its volume remains constant. And this incompressibility is great every day. This allows the hydraulic brakes of a car to stop a vehicle of several tons. The force is transmitted directly into the pipes, without any loss of volume. So, to sum up, the liquid is elusive, it flows between the fingers, it has no fixed shape, its surface always remains horizontal at rest, but, just like the solid, it is completely incompressible. Part five, the gaseous state. Often invisible, like the air you breathe, but it has incredible properties. We resume our experience with the syringe, but this time, just trapping ambient air. You mouth, you push the plunger, and there, unlike water, the air lets itself be crushed super easily. The gas is compressed in a much smaller space. And on the contrary, if you pull the plunger vigorously, the air will literally heat from the inside to fill the whole new space. It naturally expands to occupy all available space. This little test with the syringe proves three things. The gases are compressible, they are expansible, and they are elastic. If the piston is dropped, the air acts as a spring and pushes the piston back to its starting volume. It is this elasticity of air that makes bicycle tires able to absorb shocks on a damaged road. He is elusive, he has no fixed form, he has no fixed volume, because he is super compressible, and his mission in life is to occupy all the space he is given. Last part, the summary. We'll put all these discoveries together in a clear picture. Here is the perfect tool to refresh your memory. It immediately shows the commonalities and differences. Solids and liquids share incompressibility. Liquids and gases, which are grouped under the term fluid, share the fact of adapting to their containers, without having their own shape. But finally, there is only the solid that can be held in his hands, and there is only gas that agrees to change volume. Come on, finally, here's a little scientific case. Think about the sand. It flows between the fingers, it perfectly takes the shape of a beach jump. Does that

make it a liquid? Or is it just a huge collection of tiny solids, since each little grain keeps its own fixed shape? This is the kind of amazing paradox that shows that matter still has small surprises in store. Let us keep this question in mind for the next time.